

TASK 5: E-Commerce Return Rate Investigation

1. Introduction

E-commerce companies handle thousands of customer orders every day. Product returns are a major challenge because they increase logistics costs, create inventory issues, and reduce profitability. This project aims to investigate the factors influencing product returns using statistical analysis and data analytics techniques.

2. Problem Statement

The return rate of an e-commerce company has increased significantly, leading to operational and financial losses. The objective of this project is to identify the key factors contributing to product returns and provide recommendations to reduce unnecessary returns.

3.Objectives

- To analyze product return behavior using statistical techniques.
- To identify factors influencing product returns.

- To investigate the relationship between delivery delays and returns.
- To analyze the impact of product categories, discounts, and customer ratings on returns.
- To segment customers based on purchasing and return behavior.
- To provide recommendations for reducing return rates.

4. Dataset Description

The dataset contains information related to customer orders and returns. The main attributes include:

- Order_ID
- Customer_ID
- Product_Category
- Product_Price
- Discount_Percent
- Delivery_Days
- Customer_Rating
- Return_Status
- Return_Reason

The dataset was analyzed using Python, Pandas, Matplotlib, and statistical techniques.

Phase 1: Data Understanding

The dataset was loaded into Google Colab and examined for missing values and duplicate records. Basic information such as total orders, total returned products, and return percentage was calculated. Data cleaning ensured consistency and reliability for analysis.

Phase 2: Descriptive Statistical Analysis

Descriptive statistics including mean, median, standard deviation, and quartiles were computed for numerical attributes such as product price, delivery days, and customer ratings. This helped understand the distribution and variability of the data.

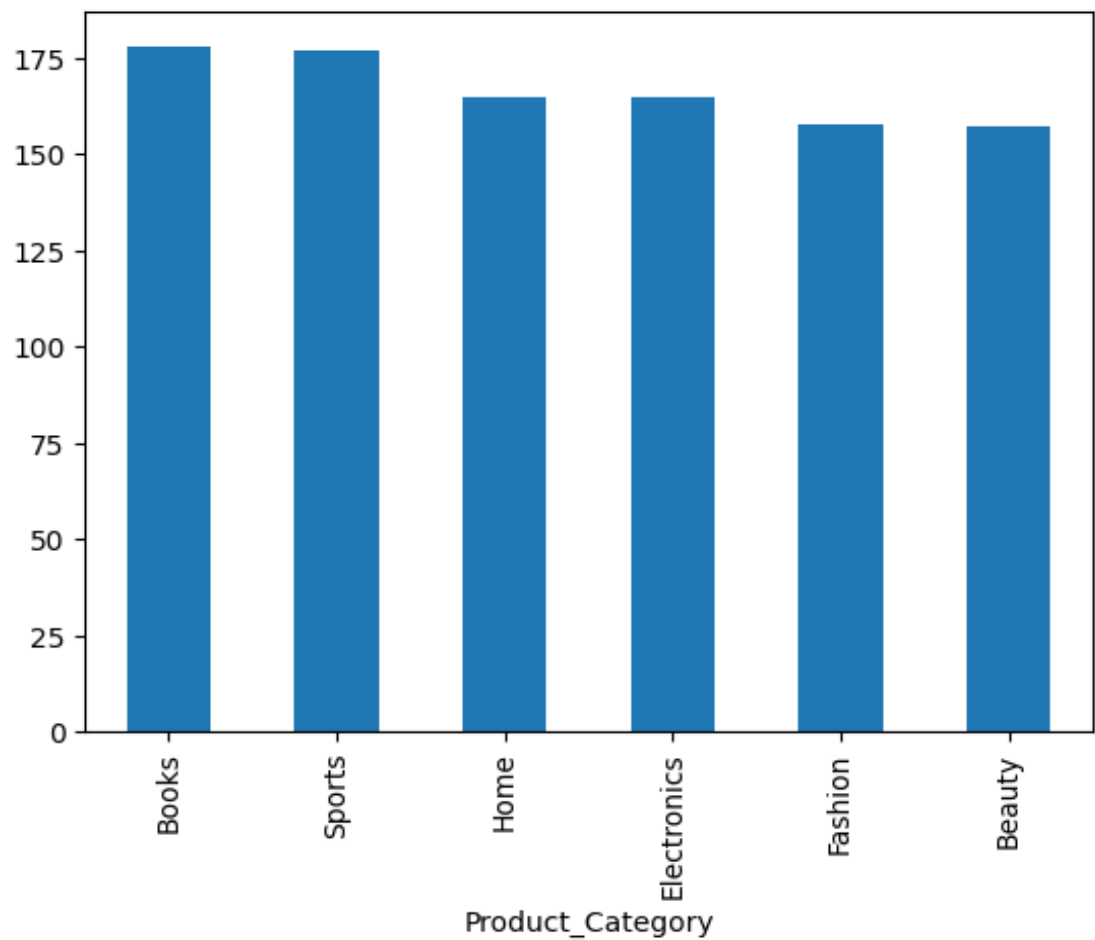
Phase 3: Return Pattern Investigation

Relationships between return status and business variables were analyzed. Product category, delivery days, discounts, product price, and customer ratings were compared against return behavior. Visualizations such as bar charts,

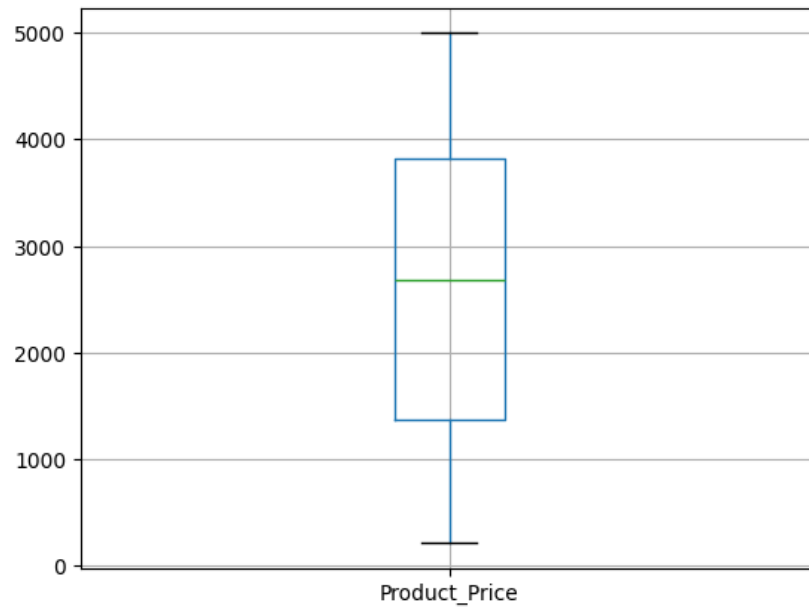
histograms, boxplots, and heatmaps were used to identify patterns and trends.

Visualizations:

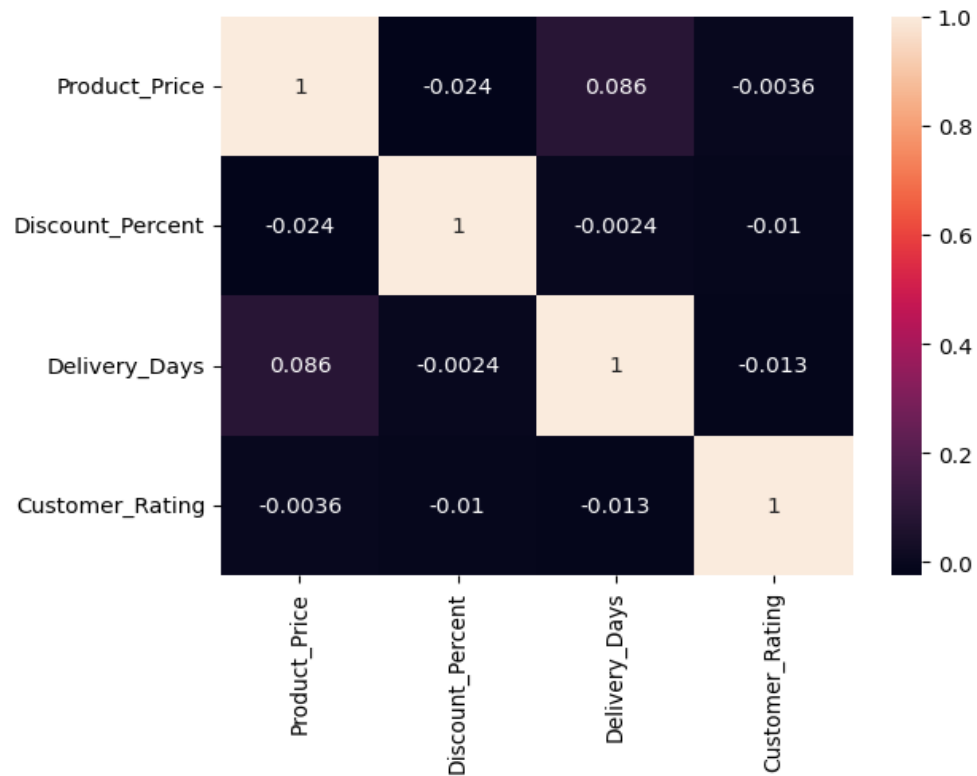
- Barcharts



- Boxplots



- Heatmaps



Phase 4: Outlier Analysis

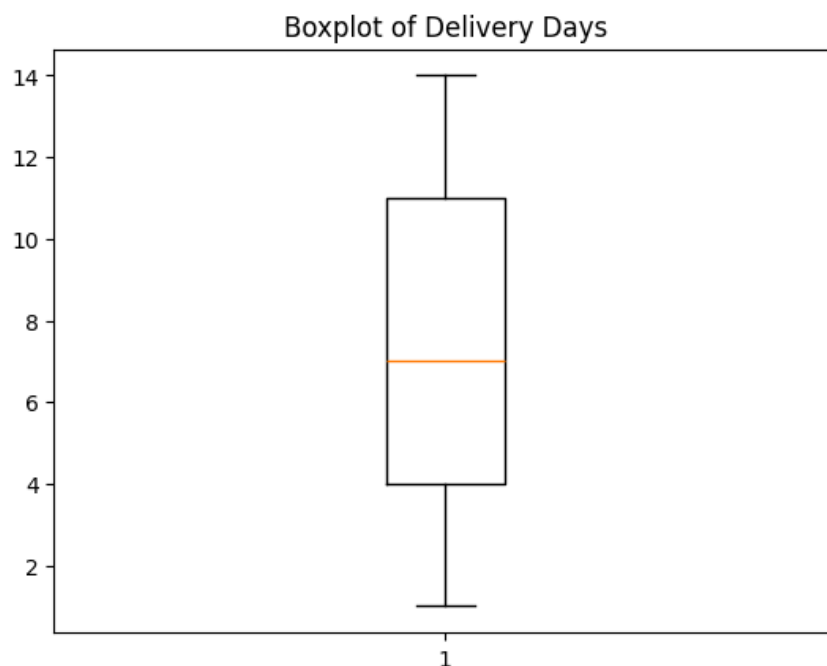
Outlier detection was performed using:

- Boxplots
- Interquartile Range (IQR)
- Z-Score Method

Outliers identified:

- Customers with unusually high return frequencies
- Extremely delayed deliveries
- Expensive products with high return rates

These outliers may indicate operational issues or special customer behavior.



Phase 5: Hypothesis Testing

A Chi-Square Test was conducted to determine whether Product Category and Return Status are associated.

Hypotheses

H_0 : Product Category and Return Status are independent.

H_1 : Product Category and Return Status are associated.

Based on the p-value obtained, the relationship between category and return behavior was evaluated.

Phase 6: Customer Segmentation

Customers were grouped into different categories based on purchasing patterns and return behavior:

- High-return customers
- Premium customers
- Discount-driven customers
- Delayed-delivery customers
- Low-rating customers

This segmentation provided deeper insights into customer behaviour.

Business Insights

1. Which factors contribute most to returns?

The analysis identified several factors that contribute significantly to product returns. Delayed deliveries were one of the major reasons, as customers often become dissatisfied when orders do not arrive on time. High discount percentages also increased return rates because customers tend to make impulsive purchases during promotional offers. In addition, products with low customer ratings showed a higher probability of being returned. Certain product categories also experienced more returns due to quality issues, size mismatches, or unmet customer expectations.

2. Which categories are risky?

Product categories with higher return rates are considered risky categories. These categories contribute a large portion of total returns and may have issues related to product quality, inaccurate descriptions, sizing problems, or customer dissatisfaction. Such categories require continuous monitoring and improved quality control measures to reduce return frequency and improve customer experience.

3. Does delayed delivery increase return probability?

Yes, delayed delivery increases the probability of product returns. Customers expect timely delivery, and delays often lead to dissatisfaction and reduced trust in the company. The analysis showed that orders with longer delivery times were more likely to be returned than orders delivered on schedule. Therefore, improving delivery performance can significantly reduce return rates.

4. Which customer groups need intervention?

The analysis identified several customer groups that require special attention. High-return customers frequently return products and contribute heavily to overall return volumes. Discount-driven customers often purchase products during sales and promotions but may later return them. Customers who experience delayed deliveries and those who provide low ratings also require intervention, as they are more likely to return products and negatively impact customer satisfaction metrics.

What actions should operations teams take?

The operations team should focus on improving delivery efficiency to ensure products reach customers on time.

Product quality checks should be strengthened before shipping to reduce defects and damages. Accurate product descriptions, specifications, and images should be provided to minimize customer misunderstandings. The company should also monitor high-return categories and customers regularly and improve customer support services to quickly resolve complaints. These actions can help reduce return rates, improve customer satisfaction, and increase overall business performance.

Recommendations

Based on the findings, the following recommendations are proposed:

Operational Improvements

- Improve delivery speed and logistics efficiency.
- Monitor delayed shipments proactively.
- Strengthen warehouse and fulfillment processes.

Product Management

- Enhance product quality assurance procedures.

- Improve product descriptions, specifications, and images.
- Conduct category-specific return analysis.

Customer Management

- Monitor high-return customers.
- Implement customer feedback programs.
- Provide proactive support for dissatisfied customers.

Pricing and Marketing

- Review discount strategies.
- Avoid excessive discount campaigns that encourage impulsive purchases.
- Develop targeted promotional offers based on customer segments.

Conclusion

This project successfully investigated the factors affecting product returns in an e-commerce environment. Statistical analysis, customer segmentation, and visualization techniques helped identify the major causes of returns. By implementing

the recommended actions, the company can reduce return rates, improve customer satisfaction, and increase operational efficiency.